

SEUNGJOON LEE

email: sjoon0503@gmail.com

LinkedIn: www.linkedin.com/in/seungjoon-lee-b455321a5

WORK EXPERIENCE

CarbonSix Robotics

Senior Software Engineer

Seoul, South Korea

Dec. 2024 - Present

- Drove the end-to-end learning, software integration, and control pipelines for dexterous, multi-modal robotic manipulation in complex manufacturing automation tasks
- Implemented advanced imitation learning algorithms and contributed to the development of SigmaKit, a complete package to collect data, train the network and deploy it on the robot to automate a task

NAVER LABS

Applied Scientist

Seoul, South Korea

Mar. 2021 - Dec. 2024

- Conceptualized and built advanced server management systems (*Garoo* and *Sero* robots) to solve complex automation problems in hyperscale data centers, including the development of path and motion planning for autonomous navigation and control algorithms for precise manipulation
- Developed optimization-based computer vision and alignment mechanisms (utilizing Ceres Solver, non-linear optimization, and AprilTags) to achieve precise object manipulation within a 1 mm tolerance
- Coordinated cross-functional alignment across electrical, mechanical, and software team members to achieve desired control objectives
- Pioneered Multi-Agent Path Finding (MAPF) frameworks bridging structural multi-agent theory with actual robot dynamics, resulting in a peer-reviewed publication at top AI conferences (ECAI 2024)

Snowbush IP

Junior Verification Engineer

Toronto, Canada

May. 2012 - Aug. 2013

- Improved and automated verification flow of digital design using various tools including Perl and HTML

EDUCATION

Seoul National University

Ph.D from the Department of Electrical and Computer Engineering

Seoul, South Korea

Sep. 2014 - Feb. 2021

- Thesis: Consensus of linear time invariant multi-agent systems over multilayer network
- Research Areas: multi-agent system, linear and nonlinear control theory, optimization, optimal control
- Supervisor: Hyungbo Shim

University of Toronto

BASc in Engineering Science, major in Energy Systems

Toronto, Canada

Sep. 2009 - May. 2014

AWARDS

- Outstanding Paper Award at ICCAS 2017.

PUBLICATIONS

The complete list of publications and citations is available at <https://scholar.google.com/citations?user=pLYkIDMAAAAJ>.

Journal Papers

1. **Seungjoon Lee** and Hyungbo Shim, "Blended dynamics approach to distributed optimization: sum convexity and convergence rate," *Automatica*, vol. 141, April 2022, doi: <https://doi.org/10.1016/j.automatica.2022.110290>.

2. Jeong Mo Seong, Jeong Woo Kim, **Seungjoon Lee**, and Hyungbo Shim, “Distributed dynamic quantile solver with plug-and-play operation,” IEEE Access, vol. 9, pp. 165517–165525, December 2021, doi: <https://doi.org/10.1109/ACCESS.2021.3134655>.
3. **Seungjoon Lee**, “Continuous-time optimization algorithm using integral control,” Journal of Institute of Control, Robotics and Systems, vol. 25, no. 9, pp.853–857, Sep 2019 (in Korean).
4. **Seungjoon Lee**, Jin Gyu Lee, and Hongkeun Kim, “Class-K function bounds for positive definition functions on compact sets,” International Journal of Control, Automation and Systems, vol. 17, no. 10, pp.2550–2555, July 2019
5. **Seungjoon Lee** and Hyungbo Shim, “Distributed algorithm for economic dispatch problem with separable losses,” in IEEE Control Systems Letters, vol. 3, no. 3, pp. 685–690, July 2019, doi: <https://doi.org/10.1109/LCSYS.2019.2916250>. (Contents of this paper were also selected by CDC 2019 Program Committee for presentation at the CDC 2019.)

Conference Papers

1. Vassilissa Lehoux, Tomi Silander, Christelle Liodice, **Seungjoon Lee**, Albert Wang, and Sofia Michel, “Multi-Agent Path Finding with Real Robot Dynamics and Interdependent Tasks for Automated Warehouses”, European Conference on Artificial Intelligence (ECAI), 2024, doi: <https://doi.org/10.3233/FAIA241017>.
2. **Seungjoon Lee** and Hyungbo Shim, “Blended dynamics approach for analysis and construction of distributed optimization algorithms,” IEEE International Conference on Control, Automation and Systems (ICCAS), 2020.
3. **Seungjoon Lee** and Hongkeun Kim, “Synchronization of Lur’e-type nonlinear systems using incremental output-feedback passivity with shortage of passivity,” IEEE International Conference on Control, Automation and Systems (ICCAS), 2020.
4. Donggil Lee, **Seungjoon Lee**, Taekyoo Kim and Hyungbo Shim, “Distributed algorithm for the network size estimation: blended dynamics approach,” in Proceeding of the IEEE Conference on Decision and Control (CDC), Miami, 2018.
5. **Seungjoon Lee** and Hyungbo Shim, “Preliminary results on distributed solution to economic dispatch problem with power losses,” in IEEE International Control, Automation and Systems (ICCAS), 2018.
6. **Seungjoon Lee**, Hyunjun Yun, and Hyungbo Shim, “Practical synchronization of heterogeneous multi-agent system using adaptive law for coupling gains,” in Proceedings of the American Control Conference (ACC), Milwaukee, 2018.
7. **Seungjoon Lee** and Hyungbo Shim, “Consensus of linear time-invariant multi-agent system over multilayer network,” in IEEE International Conference Control, Automation and Systems (ICCAS), 2017.
8. Hyunjun Yun, **Seungjoon Lee** and Hyungbo Shim, “Consensus-based distributed coordination for optimal energy generation of hierarchical systems,” in 56th Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE), 2017.
9. Jaeyong Kim, Jin Gyu Lee, **Seungjoon Lee**, Hyuntae Kim, Hyungbo Shim, Yongsoon Eun and Kangwon Lee, “Three-dimensional time-space grid allocation algorithm for traffic control of autonomous vehicles at an intersection”, in IEEE International Conference Control, Automation and Systems (ICCAS), 2015.

INVITED TALKS

1. Invited speaker. IFORS 2021. “Blended dynamics approach to distributed algorithms for the consensus optimization problem”, Aug. 25th, 2021.

SELECTED PROJECTS

Formation control of mobile robots for military applications

Sep. 2019 - Feb. 2021

- Developed a practical algorithm for formation control of mobile robots considering issues including but not limited to local reference frames, lack of orientation measurements/IMU, noise/offsets of measurements, communication/measurement delays
- Developed a tracking algorithm such that followers follow the leader while preserving formation

- Implemented the proposed algorithms into mobile robots

Motion control for vehicular dynamics

Aug. 2018 - April 2020

Sponsored by Samsung Advanced Institute of Technology

- Developed reinforcement learning algorithm for local path planning and control of the vehicle
- Integrated RL-based algorithms with model predictive control to improve safety and predictability

Distributed algorithm for power distribution in a microgrid

Sep. 2015 - Aug. 2017

Sponsored by Korea Electric Power Corporation

- Developed a distributed optimization algorithm for solving economic dispatch algorithm
- Proposed algorithm is extended to the hierarchical system

PATENTS

“The Method and System That Automatic Control of Individual Robots Using A Universal Inference Model.” <https://doi.org/10.8080/1020250091574>

“The Method and System for Process Automation Based on Imitation Learning Using a Robot Hand Apparatus for Both Teaching and Motion.” <https://doi.org/10.8080/1020250137129>

J. Cho, A. Wang, **S. Lee**. “Server Transfer Robot.” Korea patent pending (10-2023-0016552, submitted 2023/02/08); PCT pending (PCT/KR2024/001113, submitted 2024/01/24), <https://doi.org/10.8080/1020230016552>

J. Cho, **S. Lee**, A. Wang, J. Lee, J. Shin, H. Chae, J. Park. “Server Management Robot and Server Management System.” Korea patent pending (10-2023-0019474, submitted 2023/02/14); PCT pending (PCT/KR2024/001199, submitted 2024/01/25), <https://doi.org/10.8080/1020230019474>

H. Shim, J. Kim, J. Lee, **S. Lee**, H. Kim, Y. Eun, S. Son. “Non-stop intersection system and providing method thereof”, Korea Patent 10-2017-0101520, 2017, <https://doi.org/10.8080/1020160024136>.

SKILLS

Programming

C++

- 3 years of experience in developing software to control robots using C++17
- Experiences with libraries such as ROS2, Eigen, AprilTags, ceres-solver, OpenCV, abseil

python

- Basic knowledge to quickly develop scripts for a range of applications such as data gathering, analysis, visualization, and prototyping
- Experiences with libraries such as pandas, numpy, scipy, ROS2, flask, matplotlib, grpc, pytorch

Others

- linux, MATLAB, simulink, github, docker, etherCAT, real-time OS

Language

- English (proficient), Korean (fluent)

MEDIA COVERAGE

Robots in Data Center GAK Sejong

Nov. 2023

The project I contributed to received notable media coverage, a selection of which is listed below.

News Articles

- Yates, Bill. “Data Centre of the Month: GAK Sejong, Naver Technologies, South Korea.” *Capacity Media*, 30 Jan. 2024, www.capacitymedia.com/article/2creyfnroj7k4053n1ts0/news/data-centre-of-the-month-gak-sejong-naver-technologies-south-korea. Accessed 12 May 2024.

- Min-suh Ko, and Minu Kim. “Naver’s Latest Data Center Powers up - Pulse by Maeil Business News Korea.”, *Maeil Business News Korea*, 9 Nov. 2023, <https://pulsenews.co.kr/view.php?year=2023&no=863832>. Accessed 12 May 2024.
- Chea, Sarah. “Naver Bets Big on Cloud with Gak Sejong Data Center.” *Korea JoongAng Daily*, 8 Nov. 2023, <https://koreajoongangdaily.joins.com/news/2023-11-08/business/industry/Naver-bets-big-on-cloud-with-Gak-Sejong-data-center/1908570>. Accessed 12 May 2024.

Official Blog Posts

- Sero and Garo: <https://www.navercorp.com/en/storyDetail?seq=2377>
- Gak Sejong Data Center and Robots: https://www.youtube.com/watch?v=E1k661x1o_g

Videos

- Garo: <https://youtu.be/-DZfQjuevB0?si=uSds0o-kL-Sr4dgZ>
- Sero: <https://www.youtube.com/watch?v=jnrm61FY59E>
- Robostation: <https://www.youtube.com/watch?v=cLzzHE5eJt4>
- SigmaKit: <https://www.youtube.com/watch?v=0i85b3S7Yvg>